

Vadym Kurylenko

Curriculum Vitae

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Institute of Algebra and Geometry, OVGU Magdeburg

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Personal data

Date of birth October 1995

Nationality Ukrainian

Employment

1/2025 – 1/2028 **Postdoc**, *Otto-von-Guericke Universität Magdeburg*
Mentor: Benjamin NILL.

Education

- 10/2020 – 12/2024 **PhD in Geometry and Mathematical Physics**, *SISSA, Trieste*
Thesis: Local Ehrhart Theory and Gale Duality.
Supervisor: Fernando RODRIGUEZ VILLEGAS.
- 10/2019 – 9/2020 **Doctoral studies**, *Universität Hamburg*
Supervisor: Murad ALIM.
- 10/2017 – 9/2019 **Master's Degree in Mathematical physics**, *Universität Hamburg*
Thesis: Gauss-Manin Lie Algebra for Non-Compact Calabi-Yau Manifolds
Final mark: 1.1/1.
Supervisor: Murad ALIM.
- 9/2013 – 9/2017 **Bachelor's Degree in Physics**, *V.N. Karazin Kharkiv National University*
Thesis: The Regge-Wheeler Equation in a Quasi-Spherical Neutral Black Hole Background.
Final mark: 88/100, with honours.
Supervisor: Alexei NURMAGAMBETOV.

Publications and preprints

Publications

- 2026 **Thin Simplices via Modular Arithmetic**,
to appear in Combinatorial Theory, <https://arxiv.org/abs/2404.03975>
- 2022 **The algebra of derivations of quasi-modular forms from mirror symmetry**,
with M. Alim and M. Vognin, <https://dx.doi.org/10.4310/PAMQ.2022.v18.n3.a6>

Preprints

- 2026 **The Euler Stratification for $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^n$** , with S. Hoşten, E. Neuhaus and N. Rieke, <https://arxiv.org/abs/2603.19184>
- 2026 **Preserving Hodge Vectors of Lattice Polytopes**, with B. Nill, <https://arxiv.org/abs/2602.20765>
- 2025 **Examples of IDP lattice polytopes with non-log-concave h^* -vector**, with J. Hofscheier and B. Nill, <https://arxiv.org/abs/2505.18896>

Talks

- 05/2026 **Seminar on Nonlinear Algebra**, *Leipzig*, (invited)
Preserving Hodge Vectors of Lattice Polytopes
- 9/2025 **Ukrainian Summer School in Combinatorics**, *Freiberg*, (contributed)
Thin Lattice Simplices
- 9/2025 **Modern Perspectives on Geometry of Numbers**, *Cottbus*, (invited)
Thin Lattice Simplices
- 3/2025 **Combinatorial Synergies East**, *MPI, Leipzig*, (invited)
Local Ehrhart Theory and Its Synergies
- 2/2025 **Workshop on Polytopes**, *Universität Osnabrück*, (contributed)
Ehrhart Theory and Gale duality
- 7/2024 **Conference on Arithmetic Geometry and Applications**, *ICTP*, (contributed)
Ehrhart Theory and Gale Transforms
- 9/2023 **Workshop "Hodge and K-theory meet combinatorics"**, *EPFL*, (contributed)
Thin simplices via modular arithmetic.
- 5/2023 **Oberseminar in Magdeburg**, *OvGU Magdeburg*, (invited)
Thin simplices via modular arithmetic.
- 4/2023 **Graduate Student Meeting on Applied Algebra and Combinatorics**, *KTH, Stockholm*, (contributed)
Vanishing of local h^* -polynomial for simplices.

Awards and Scholarships

- 10/2020 – 9/2024 **SISSA Fellowship**, 4-year fully funded scholarship
- 10/2017 – 9/2019 **DAAD Study Scholarship for Graduates of All Disciplines**, Scholarship for Master studies in Germany, c.100 recipients from Ukraine yearly.
- 2015 – 2017 **Scholarship of Verhovna Rada (Parliament) of Ukraine**, one nominee from the department and 500 scholarships to be awarded each year nationwide.

Visits

- 06/2023 **OvGU Magdeburg**, *Magdeburg*, Germany
Visitor as a guest of Prof. Benjamin Nill

Teaching

- Spring 2022 ICTP, Teaching assistant for Representation Theory
- 2014-2016 Coaching a school team for Regional Olympiads in Physics and Astronomy in Kharkiv

Languages and Computer Skills

- Languages Russian (Native), Ukrainian (Native), English (Fluent), German (Level A2.2), Italian (Level A1).
- Computer Skills Sagemath, Python; familiarity with Polymake, Mathematica and Magma.